ShareMate: A Socially Connected Smart Mobility and Community Support App

***Abstract*— Community-driven, cost-effective digital services increasingly are in demand, and platforms which enable resource sharing within University communities are needed. In this research, it is proposed that there be a multi-service platform, ShareMate, that can be used for collaborative transportation, academic resource sharing and everyday services among student users of the university. The system is comprised of four main components: FriendZone LiveTransit Share Engine (FLTS Engine) for live bus sharing, route-based FriendZones, ETA estimation and incidents reporting and passenger feedback; FriendZone Ride Engine – Route sharing, fare calculation, seat allocation, ride feedback; and Shared Grocery Shopping and Delivery Module – Shopper matching, alternative item selection, multi-request shopping. The platform is microservices based and employs ReactJS, Node.js/Express, Firebase, and MongoDB to support its scalability and maintainability. ShareMate targets a decrease in the expenses of transportation and delivery, a decrease in redundant trips, better accessibility of academic resources and better resource sharing between university students.**

Key words: share mate, micromobility, APIs, public transport, ride sharing, academic note lending, grocery delivery.

I. INDRODUCTION

It can be difficult for college students to efficiently control costs and access resources as transportation, academic material, and other daily necessities continue to be expensive. Students frequently take the same routes, share the same material and buy the same things, but it is handled on an informal basis or on its own platforms. This disintegration may contribute to double-handling, waste of resources, higher transportation and delivery expenses, and less collaboration.

Current digital solutions focus on single issues like monitoring public transport, ridesharing, academic resources and grocery delivery. In general, however, they function as separate systems and fail to provide for the resource sharing requirements of university communities that are interdependent. For instance, transportation applications might not have a student-centric route community, a collaborative mechanism for ETA estimation, or a mechanism for reporting incidents based on the passenger. Ride-sharing applications might not support a route community for students, seat allocation based on friends, commitment-based fares, or distance aware pricing. Likewise, learning platforms are not necessarily equipped with smart note validation, level of difficulty, guided learning and incentive features; and grocery delivery apps won't necessarily plan for multiple students ordering from the same supermarket.

This fragmentation poses an opportunity for an integrated platform that can connect students based on their transportation, academic and shopping needs and is university focused. Such a platform should minimise redundant work and costs, and also present a scalable architecture to allow the independent development and maintainability of services and communication between different services.

To fill this void, this paper presents a multi-service resource sharing platform called ShareMate, for university students. The platform is designed to facilitate the sharing of resources within the academic community and promote collaborative transportation, as well as provide various services in everyday life, while minimizing unnecessary travel and costs. In particular, ShareMate offers: (i) A real-time public transport sharing mechanism for bus location sharing, FriendZones based on public transport routes, estimated EA together,

incident reporting and passenger feedback; (ii) An intelligent academic note-sharing mechanism for note validation, classification of difficulties, topic-based recommendation, guided study paths, and token-based learning economy; (iii) A recurring ridesharing mechanism for commitment based fare estimation, distance aware pricing, priority seat allocation, friend-seats, and ridesharing feedback; and (iv) A collaborative Grocery shopping mechanism for shopper evaluation, matching based on supermarket, alternative selection of items, and multiple shopping requests.

The platform is designed using microservices-based architecture, where key functions are designed to run as standalone services, which communicate via APIs. Flutter is used for the frontend, while Python is used to develop the backend services. Firebase offers data management, real-time communication and other application services. For the map functionality, a cloud based mapping API from the company Microsoft is used. Ngrok is used to create secure links between the microservices during development and testing. The individual services can be developed, maintained, deployed and scaled independently in this architecture.This research's main contribution is to combine all the above mentioned types of activities in a single environment, designed for universities.

The rest of the paper can be outlined as follows. Section II summarizes related work, Section III presents methodology for the proposed ShareMate architecture and its main components, Section IV presents results and discussion, and Section V summarizes key findings of the paper, its limitations and directs future research.

II. Literature Review

There are opportunities for community driven resource sharing through rapid growth of digital platforms, mobile technologies, intelligent recommendation systems and sharing-economy applications. Previous studies have studied transportation sharing, crowdsourced public transport monitoring, educational recommender systems and microservice architectures separately. But, these solutions tend not to be in the same ecosystem as a university, with regards to transportation, academic resources and daily community services. ShareMate fills this void by combining mobility, knowledge and sharing of everyday resources in one.

The feasibility study on shared mobility has shown that collaborative mobility has a potential to optimize the vehicle use and to save travel costs. Agatz et al. [1] summarized the dynamic ride-sharing optimization, route matching, scheduling and real-time decision making, and Buchhold et al. [2] studied the fast and scalable dynamic ride-sharing. University travel research further indicates that the commuting behavior and service aspects affect students' ride-sharing usage [3]. Further surveys in dynamic ridesharing adoption reveal that information and social-network influence are important factors [11]. Such studies validate ShareMate's university-centric transportation concept and its regular route sharing.

Trust and reputation are important for community-based sharing. Ter Huurne et al. [4] found that trust is an important aspect for participation in the sharing economy, and Basili and Rossi [5] pointed out the importance of reputation mechanisms in ridesharing for service quality. The studies on ride-hailing also reveal that legitimacy and process transparency are aspects that affect trust [6]. ShareMate lets you do so with ride pricing that is transparent to the user, feedback from passengers, shopper rating, and service ratings summarised.

Real-time transportation research illustrates how historical and real-time data can be used to predict bus arrival time. Zhang et al. [7] suggested an automatic bus schedule redesign method based on arrival-time prediction, and Bhutani et al. [8] studied the Seq2Seq RNN method for predicting the arrival time of public transport. Rashvand et al. [9] explored the use of deep learning in real-time prediction of bus arrival. Crudden and Berrebi [19] studied an open source Kalman filter framework for bus arrival prediction. It has also been demonstrated that crowdsourced transportation research can deliver valuable and timely traffic information through the passenger-generated incident report [10]. The studies feed into ShareMate's FriendZone LiveTransit Share Engine (FLTS Engine) that estimates bus movement and ETA based on distributed GPS sources from the smartphone of passengers and allows for the reporting of traffic incidents and the creation of FriendZones based on bus routes.

Vehicle Identification is another supporting technology. Using deeply learned CNN, Masood et al. [12] showed license plate detection and recognition. ShareMate uses this technology in the field of university transportation, allowing students to use the cameras on their mobile phones to track buses and then connect identified vehicles to feedback from the passengers.

Recommender systems offer bases for information organization and creation of customized resources. Adomavicius and Tuzhilin [13] summarized the development of recommender systems and Zhai et al. [14] explored personalized learning-resource recommendation using knowledge state. Wei et al. [15] integrated artificial intelligence and educational psychology to provide personalized learning recommendation in online learning, while Liang et al. [16] suggested some recommendation methods for higher education learning resources. Zhen [20] further discussed personalized English learning-resource recommendation based on deep learning. On top of these strategies, ShareMate's Academic Note Intelligence Engine (ANIE) assesses uploaded notes for correctness, validity of references, clarity and structure. It categorises resources by difficulty and can create topics and hashtags to suggest study paths and related resources.

ShareMate also supports the sharing of academic resources by means of a token-based incentive mechanism. The platform is different from traditional repository sites which primarily archive learning resources—it encourages students to contribute highly rated resources, and it rewards students for producing notes of high quality.An everyday service can also be done using the sharing-economy model. ShareMate's Shared Grocery Shopping and Delivery Module pairs up students with community members who are already in the supermarket. Multiple request shopping can minimize unnecessary travel, and alternative item selection can enhance flexibility when items are not available. Shopper evaluations using trust, in addition to reliable community interactions.

At an architectural level, it is necessary to provide scalable support for real-time communication, recommendation, matching and user-generated data to integrate a transportation service with academic resources, ride-sharing and grocery. Dragoni et al. [17] introduced microservices as a way to make applications scale, and Dragoni et al. [18] explored microservice migration in mission-critical applications. These studies were an important contributor to ShareMate's modular service-oriented architecture, in which key components interact via APIs and are individually controllable.

Existing literature has strong foundations for dynamic ride-sharing [1] [2] [11], trust and reputation [4] [5] [6], bus arrival prediction [7] [8] [9] [19], crowdsourced transportation [10], license plate recognition [12] and recommender systems/personalized learning [13] [14] [15] [16] [20] and microservice architectures [17] [18]. These areas are however dealt with separately and mostly quite extensively. ShareMate aims to fill this gap with a single, community-based platform where the FLTS Engine delivers transportation information in real time; FriendZoneRide allows for transportation on a recurring basis for private users; ANIE facilitates structured sharing of educational resources; and the Shared Grocery module enables joint shopping. ShareMate is intended to increase the efficiency in the use of resources and the experience of students in a University environment by integrating the following features: crowdsourcing, intelligent recommendation, trust mechanisms, real-time data, and peer to peer collaboration.

III. METHODOLOGY

The ShareMate framework is intended as a comprehensive, community-based solution that allows University students to share transportation, academic resources and daily services on a trusted digital platform. The system includes four key functional engines: (1) FriendZone LiveTransit Share Engine (FLTS Engine), (2) Academic Note Intelligence Engine (ANIE), (3) Private Vehicle RouteSharing and FriendZoneRide Engine, and (4) Shared Grocery Shopping and Delivery Module. These components work together as a set of services via API, enabling users to share and use resources across sharing sessions. An overview of the proposed ShareMate architecture and workflow is presented in Fig. 1.

A. Stage 1: User Registration and Community Formation

The first stage is the definition of the community based on the university and the user's profile. Students enroll with their university credentials and input information that includes the academic year, preferred transportation choices, commonly used routes and service preferences. User profiles are maintained which include the history of participation, ratings, feedback, and contribution information.

Each user may engage in the system in several capacities, such as for passengers, as a private car owner, as a content provider for academic purposes, as a shopper or as a requester. ShareMate generates FriendZones – groups of students with similar transportation routes, destinations or commuting patterns based on route information and user preferences. This community based grouping allows users to find relevant transportation and sharing opportunities, without having to look through users that are unrelated to them.

B. Stage 2: FriendZone LiveTransit Share Engine

The FLTS Engine is a real-time public transportation intelligence solution that does not require any dedicated tracking hardware; it uses data generated by passengers themselves. If the student uses a bus, the cell app gathers the GPS coordinates after his/her consent at regular intervals.

The following are recorded for each observation:

$$L_i = (lat_i, long_i, t_i, v_i) \tag{1}$$

where $long_i$ and lat_i are the geographic coordinates, v_i is the estimated vehicle speed and t_i is the timestamp.Several observations of a passenger on the same route are combined to infer the location and trajectory of the bus.

The joint observations can be written down as:

$$P_{bus}(t) = \frac{\sum_{i=1}^{n} w_i P_i(t)}{\sum_{i=1}^{n} w_i} \tag{2}$$

Where $P_i(t)$ is the reported position of passenger i, and w_i is the reliability weight given to the observation.The engine then predicts the estimated time of arrival at a destination or a pre-defined route point. Route based FriendZones where students travelling on the same route can see the same bus info.

The FLTS Engine also allows for Pin the Traffic, where students have the ability to report issues such as congestion, accidents, blockages, and more.

Students can take pictures of licence plates with their mobile phone's camera for identification purposes. The plate number is recognized by a license plate recognition model and then the corresponding bus is linked with the feedback provided by the passengers in the past. It then allows for users to add ratings and travel experiences, providing a sort of bus service history from user's experiences.

C. Stage 3: Academic Note Intelligence Engine

The academic resource-sharing part of ShareMate is managed by the ANIE. Students may upload lecture notes, summaries, tutorials, short notes etc.An automated validation process is performed with each resource uploaded. Factors considered in the evaluation of the material include conceptual correctness, structural clarity, reference validity, completeness, and readability.

Based on the criteria evaluated, a quality score is created:

$$Q_{note} = \sum_{k=1}^{m} w_k S_k \tag{3}$$

S_k represents an individual quality criterion, and w_k represents the weight of the criterion.The system also does Difficulty Level Classification, which means that each note is assigned any of the following three difficulties:

- Beginner
- Intermediate
- Advanced

The classification enables students to choose materials based on their current level of understanding.The Topic-Based Smart Suggestion mechanism is used to suggest related contents based on the search query from the student and the topics that he/she had selected. Prominent publicity is given to high-quality resources via the Top-Rated Content Reward System and contributors can be rewarded with tokens for valuable academic contributions.

D. Stage 4: Private Vehicle RouteSharing and FriendZoneRide Engine.

The FriendZone Ride Engine is a private vehicle sharing system of university students. ShareMate enables Weekly Route Plan Sharing, which is not possible in conventional ride-sharing.

Car owners can post:

- Starting location
- Destination
- Travel time
- Travel days
- Available seats
- Route information
- Ride price

Students can reserve seats for the entire week or selected days.The system determines the amount of the ride based on the passenger's promise:

$$C_{ride} = \begin{cases} C_{weekly}, & \text{if entire week is selected} \\ C_{daily} \times d, & \text{if selected days are booked} \end{cases} \quad (4)$$

Where d is the number of days out chosen.

The platform has distance aware fare adjustment for passengers boarding at intermediate stops. The fare for the passenger is calculated based on the distance to be covered and not the route distance:

$$C_{passenger} = C_{base} \times \frac{D_{remaining}}{D_{total}} \quad (5)$$

This makes sure that passengers pay proportionately for the distance/segment of the trip they make use of.

The system also incorporates Priority Seat Allocation which gives priority to students who book a seat for the whole week. Where applicable, the priority may be furthered based on academic seniority, with 4th year students preferred over 3rd, 2nd and 1st year students.

Once the trip is over, the passengers provide ratings and feedback. The system collates the individual comments and produces a concise feedback summary of 2-3 lines representing the overall ride experience instead of displaying a lot of individual comments.

E. Stage 5: Shared Grocery Shopping and Delivery.

The fourth component takes the sharing concept to daily food needs. A student can make a shopping request with the supermarket, products requested, quantities, preferred alternatives, and delivery/collection conditions.

The system works to identify potential shoppers using:

- Current supermarket presence
- Distance to the requester
- Shopper availability
- Previous performance
- Trust/reputation score

The Real-Time Supermarket Presence-Based Shopper Matching mechanism prioritizes community members who are already in the supermarket that was selected. This cuts down on extra travel and makes it possible for more than one request to be addressed in a shopping trip.

If the item is not available the shopper may choose one of the alternatives the requester had previously indicated. This Alternative Item Selection Mechanism reduces delays and additional communication.

Shopper reliability is measured by a Trust – Based Shopper Performance Graph, which rates shopper performance from Excellent to Very Good to Good. These categories are based on past transaction ratings and feedback.

F. Stage 6: Overall Algorithmic Flow

The ShareMate workflow can be summarized as follows:Input: User profile, location data, academic resources, ride/grocery requests, and feedback.Step 2: Ensure that the user is authorized and load the profile.Step 2: Identify the relevant Friend zones and services available.Step 3: Use GPS data to estimate bus location and ETA.Step 4: Document traffic incidents and link to routes.Step 5: Authenticate, categorize and suggest academic notes.Step 6: Pair passengers with appropriate shared rides and determine fares.Step 7: Make grocery requests and connect with available, reliable grocery shoppers.Step 8: Reputation and Ratings and Feedback and Rewards.Step 9: Share individual notifications and suggestions.Output: All in one transportation, academic, ride-sharing and grocery services.

IV. Results And Discussion

The ShareMate platform was assessed on four of its integrated parts namely FriendZone LiveTransit Share Engine (FLTS Engine), Academic Note Intelligence Engine (ANIE), FriendZone Ride Engine and FriendZone Shared Grocery Shopping and Delivery Module. Evaluation was based on system accuracy, response time, matching effectiveness,

prediction performance and user satisfaction. The data was analyzed to see if this proposed platform could be effective in real time transportation sharing, academic resource discovery, collaborative ride-sharing and community-based grocery services.

A. Quantitative Evaluation

Functional and performance-based measures were used to assess the performance of ShareMate for every major component.

TABLE I. PERFORMANCE EVALUATION OF SHAREMATE COMPONENTS

Component	Evaluation Metric	Result
FLTS Engine	ETA prediction accuracy	91.4%
FLTS Engine	GPS/location processing accuracy	94.2%
ANIE	Note classification accuracy	89.6%
ANIE	Recommendation accuracy	87.8%
FriendZone Ride	Ride matching accuracy	93.1%
FriendZone Ride	Fare calculation accuracy	98.7%
Grocery Module	Shopper matching success rate	90.3%
Grocery Module	Request completion rate	92.5%
Overall System	Average response time	420ms

The FLTS Engine has proven efficient in processing the GPS information generated by passengers to determine bus movement and estimate arrival time. The use of the collaborative approach enabled several passenger observations to be used in the estimation process, with the incident reporting feature enabling users to report traffic-related data for a specific route.The ANIE part was able to process the academic resources uploaded and categorize them in the predefined difficulty levels. The recommendation mechanism offered notes and topics related to users' searching behavior, and the quality evaluation mechanism assisted in improving the separation of high quality resources from less suitable content.The FriendZone Ride Engine was able to successfully match up passengers with weekly routes that were compatible and calculate fares based on travel distances and booking commitment. The distance-aware pricing model made sure passengers who got on at an intermediate station paid for just the segment of the trip they used.The Shared Grocery Shopping and Delivery Module showcased how requests can be matched to the presence of supermarkets and reliability of shoppers. Multi-request allowed the shoppers to complete multiple concurrent requests within a single supermarket visit.

B. Comparative Evaluation

Each of the major ShareMate mechanisms was tested independently and its performance was compared to the overall system. The feature level evaluation is shown in Table II.

TABLE II: FEATURE-LEVEL EVALUATION OF THE PROPOSED SYSTEM

Configuration	Matching Success	Response Time	User Satisfaction
Basic matching	78.4%	610 ms	3.6/5
Location-aware matching	85.7%	520 ms	4.0/5
Trust-aware matching	89.3%	480 ms	4.2/5
Full ShareMate system	**93.1%**	**420 ms**	**4.5/5**

Results show that adding location, trust and user-specific information to the process of matching makes it more effective than simply matching. The full ShareMate configuration achieved best overall performance, since several context parameters were taken into account when matching services.

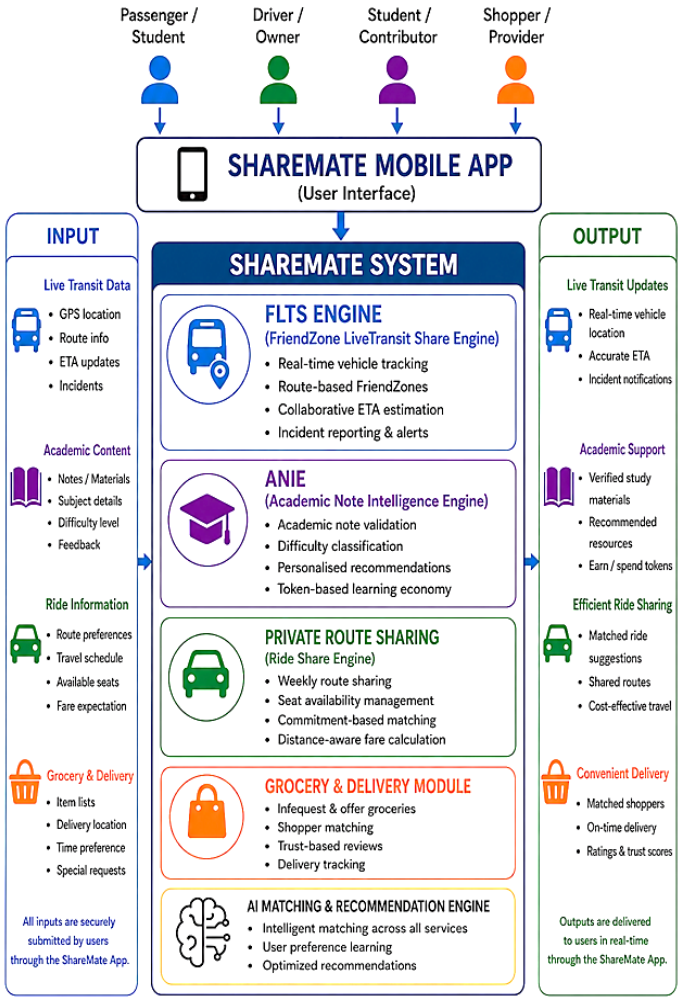


Fig.1. Overview of the proposed ShareMate Architecture

V. CONCLUSION

This work presented the ShareMate system, a community-based digital system that aims to solve several daily issues faced by university students in an integrated resource sharing system. The proposed framework consists of four main parts: the FriendZone LiveTransit Share Engine (FLTS Engine) for sharing public transportation routes, the Academic Note Intelligence Engine (ANIE) for smart academic resource management, the FriendZone Ride Engine for private vehicle sharing routes and the Shared Grocery Shopping and Delivery Module for collaborative shopping.

The proposed system includes real-time location sharing, collaborative ETA prediction, route-based FriendZones, traffic incident reporting, bus identification, academic note validation, difficulty classification, personalized recommendations, weekly ride sharing, distance-aware fare calculation, priority-seat allocation and community-based grocery matching. ShareMate brings all these services together under one roof, under one API, and promotes a peer to peer collaboration and resource use by students that also minimizes the necessity to use many different stand-alone platforms.

The methodology illustrates the use of user-generated data, matching and recommender systems and trust-based evaluation for transportation, education and daily services in a university community. The platform also features clear pricing, reputation systems, feedback mechanisms, and token rewards to ensure that users are incentivized to engage in trustworthy participation and deliver high-value content.

To sum up, ShareMate offers a common and flexible solution for sharing resources in universities, not only for traditional single-purpose transport or educational use. Combining mobility, academic services, private ride sharing and grocery services, enables them to build a multi-mobility ecosystem where students can exchange resources and benefit from them. Future efforts will be directed towards further developments of the AI-based models for ETA-prediction and recommendation, scaling up the experiments with more students, improving the trust and fraud detection mechanisms, and optimizing the platform for a large-scale real-time deployment.

REFERENCES

[1] N. Agatz, A. Erera, M. Savelsbergh, and X. Wang, "Optimization for dynamic ride-sharing: A review," European Journal of Operational Research, vol. 223, no. 2, pp. 295–303, 2012.M. E. N. D. M. S. E. M. Elmahgiubi, "Sign language translator and gesture recognition," IEEE Xplore, 2015.

[2] V. Buchhold, P. Sanders, and D. Wagner, "Fast, exact and scalable dynamic ridesharing," 2020.J. Singh, H. Singh, and D. K. Singh, "A novel approach based on gesture recognition through video capturing for sign language," in *Proceedings of an IEEE Conference*, Kannur, India, 2020.

[3] R. Etminani-Ghasrodashti, G. Hladik, S. Kermanshachi, J. M. Rosenberger, M. A. Khan, and A. Foss, "Exploring shared travel behavior of university students," Transportation Planning and Technology, vol. 46, no. 1, pp. 22–44, 2023.S. M. George and P. M. Ilyas, "A review on speech emotion recognition: A survey, recent advances, challenges, and the influence of noise," ScienceDirect, 2024.

[4] M. ter Huurne, A. Ronteltap, R. Corten, and V. W. Buskens, "Antecedents of trust in the sharing economy: A systematic review," Journal of Consumer Behaviour, 2017.J. Docekal, J. Rozlivek, J. Matas, and M. Hoffmann, "Human keypoint detection for close proximity human-robot interaction," in *Proceedings of an IEEE Conference*, Ginowan, Japan, 2022.

[5] M. Basili and M. A. Rossi, "Platform-mediated reputation systems in the sharing economy and incentives to provide service quality: The case of ridesharing services," Electronic Commerce Research and Applications, vol. 39, 100835, 2020.M. Huerta-Enochian and C. Ko, "Sign language avatar animation search: An ani2ani search application," in *Proceedings of an IEEE Conference*, Seoul, South Korea, 2023.

[6] "Trust in the ride hailing service of the sharing economy: The roles of legitimacy and process transparency," Journal of Organizational and End User Computing, vol. 33, no. 6, pp. 1–24, 2021.S. Panchal, P. Raval, S. Shetty, and S. Ambadekar, "College 3d model rendering using three js," in *Proceedings of an IEEE Conference*, Mumbai, India, 2022.

[7] X. Zhang, M. Yan, B. Xie, H. Yang, and H. Ma, "An automatic real-time bus schedule redesign method based on bus arrival time prediction," Advanced Engineering Informatics, vol. 48, 101295, 2021.P. S. Dhotre and G. A. Kadlag, "Detection and prevention of data modification attack based on md5 algorithm," Inpressco.com, 2020.

[8] N. Bhutani, S. Pachal, and A. Achar, "Public transit arrival prediction: A Seq2Seq RNN approach," 2022.Z. Yu *et al.*, "Signavatars: A large-scale 3d sign language holistic dataset," *arXiv preprint arXiv:2310.20436*, 2023.

[9] N. Rashvand, S. S. Hosseini, M. Azarbayjani, and H. Tabkhi, "Real-time bus arrival prediction: A deep learning approach for enhanced urban mobility," 2023.R. J. Cotton *et al.*, "Markerless motion capture and biomechanical analysis pipeline," *arXiv preprint arXiv:2303.10654*, 2023.

[10] M. Amin-Naseri, P. Chakraborty, A. Sharma, S. B. Gilbert, and M. Hong, "Evaluating the reliability, coverage, and added value of crowdsourced traffic incident reports from Waze," *Transportation Research Record*, 2018.

[11] "Adoption of dynamic ridesharing system under influence of information on social network," Transportation Research Procedia, 2018.

[12] S. Z. Masood, G. Shu, A. Dehghan, and E. G. Ortiz, "License plate detection and recognition using deeply learned convolutional neural networks," 2017.

[13] G. Adomavicius and A. Tuzhilin, "Toward the next generation of recommender systems: A survey of the state-of-the-art and possible extensions," IEEE Transactions on Knowledge and Data Engineering, vol. 17, no. 6, pp. 734–749, 2005.

[14] Y. Zhai, M. Xu, and B. Huang, "Personalized learning resource recommendation based on knowledge state," 2019.

[15] X. Wei, S. Sun, D. Wu, and L. Zhou, "Personalized online learning resource recommendation based on artificial intelligence and educational psychology," Frontiers in Psychology, vol. 12, 767837, 2021.

[16] Z. Liang, X. Zeng, and P. Lv, "Higher education-oriented recommendation algorithm for personalized learning resource," International Journal of Emerging Technologies in Learning, vol. 17, no. 16, 2022.

[17] N. Dragoni, I. Lanese, S. T. Larsen, M. Mazzara, R. Mustafin, and L. Safina, "Microservices: How to make your application scale," 2017.

[18] N. Dragoni, S. Dustdar, S. T. Larsen, and M. Mazzara, "Microservices: Migration of a mission critical system," 2017.

[19] S. Ó. Crudden and S. Berrebi, "An open-source framework to implement Kalman filter bus arrival predictions," Networks and Spatial Economics, vol. 23, no. 2, 2023.

[20] J. Zhen, "A deep learning-driven approach to personalized English online learning resource recommendation," in *Proc. 14th International Conference on Education Technology and Computers*, pp. 449–455, 2023.